

QUBO

$$\min x^T Q x \text{ s.t. } x \in \{0, 1\}^p$$

Given the Example in [Binary LP.md]:

$$\min x_1^2 + x_2^2 + 5x_1x_2 \text{ s.t. } x_1, x_2 \in \{0, 1\}$$

$$\equiv x_1 + x_2 + 5x_1x_2 \text{ s.t. } x_1, x_2 \in \{0, 1\}$$

QUBO is a special case of linear programming because we can represent the multiplication by other variables.

Note that Quantum Computers are REALLY good at solving these QUBO problems.

Lagrangian Relaxtion in QUBO

$$L(\lambda) = \min c^T x - \lambda(f^T x - g)$$

Boolean Condition	Linear Constraint	Quadratic Constraints
$x \vee y$	$x + y \geq 1$	$1 - x - y + xy = 0$
$x \vee y \vee z$	$x + y + z \geq 1$	$1 - x - y - z + xy + xz + yz - xyz = 0$
$x \rightarrow y$	$x \leq y$	$x - xy = 0$
$\neg(x \wedge y)$	$x + y \leq 1$	$xy = 0$
$x \wedge y$	$x + y = 2$	$1 - xy = 0$
$x \oplus y$	$x + y = 1$	$(1 - x - y)^2 =$

$$\min x^T Q x \text{ s.t. } x \in \{0, 1\}^q$$

$$= q_{11}x_1^2 + q_{12}x_1x_2 + q_{21}x_1x_2 + \dots$$

$$= q_{11}x_1 + q_{12}x_1x_2 + q_{21}x_1x_2 + \dots$$

Vertex Cover

$$\min \sum_{i \in V} x_i$$

s.t. $x_i + x_j \geq 1, (\{i, j\} \in E)$ -> this constraint is a problem so apply lagrangian Relaxtion

$$x_i \in \{0, 1\}, (i \in V)$$

Lagrangian Relaxtion

$$\begin{aligned} & \min \sum x_i + \lambda \sum_{i,j \in E} (1 - x_i - x_j + x_i x_j) \\ &= \sum_{i \in V} x_i (1 - \lambda \deg(i)) + \lambda \sum_{i,j \in E} x_i x_j \end{aligned}$$